and thus bulge out toward the Moon. The oceans on the opposite side of the Earth are pulled most weakly and thus bulge out away from the Moon. The result is two oceanic bulges; and as the Earth turns, those bulges show up as two high tides every twenty-four hours. In honor of those tides, you recall, the head-to-foot gravitational force that you felt is called a *tidal force*. You also recall that Einstein's general relativity describes this tidal force as due to a curvature of space and warpage of time, or, in Einstein's language, a *curvature of spacetime*.[23] Tidal forces and spacetime distortions go hand in hand; one always accompanies the other, though in the case of ocean tides the distortion of spacetime is so tiny that it can be measured only with extremely precise instruments.

But what about Arnold? Why was he so blithely immune to the hole's tidal force? For two reasons, DAWN explains: first, because he was much smaller than you, only 10 centimeters high, and the tidal force, being the difference between the gravitational pulls at his head and his feet, was correspondingly smaller; and second, because he was made of a superstrong titanium alloy that could withstand the stretching force far better than your bones and flesh.

Then with horror you realize that, as he fell through the horizon and on in toward the singularity, Arnold must have felt the tidal force rise up in strength until even his superstrong titanium body could not resist it. Less than 0.0002 second after crossing the horizon, his disintegrating,stretching body must have neared the hole's central singularity. There, you recall from your study of general relativity back on Earth, the hole's tidal forces must come to life, dancing a chaotic dance, stretching Arnold's remains first in this direction, then in that, then in another, faster and faster, stronger and stronger, until even the individual atoms of which he was made are distorted beyond all recognition. That, in fact, is one essence of the singularity: It is a region where chaotically oscillating spacetime curvature creates enormous, chaotic tidal forces.[24]

Musing over the history of black-hole research, you recall that in 1965 the British physicist Roger Penrose used general relativity's description of the laws of physics to prove that a singularity must reside inside every black hole, and in 1969 the Russian troika of Lifshitz, Khalatnikov, and Belinsky used it to deduce that very near the singularity, tidal gravity must oscillate chaotically, like taffy being pulled first this way and then that by a